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Research Methodology – Worksheet - 1 [Summary]

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1.) **Research:**

Research is a systematic and scientific process of finding information, solving problems, and discovering knowledge using evidence.

Research Process:

Problem Identification → Data Collection → Analysis → Interpretation → Conclusion.

Objectives:

Find answers, explain relationships, test theories, and solve real – world problems.

2.) **Types of Research**

Type	Meaning
Exploratory	Discover new ideas and patterns.
Descriptive	Describe facts and characteristics
Analytical	Analyze data and relationships.
Theoretical	Models, logic, and proofs.
Empirical	Observation and experiments.
Fundamental	Develop basic knowledge.
Applied	Solve practical problems.

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Theoretical : Prove an algorithm mathematically.

Empirical : Test a model using a dataset

3.) Research Problem:
A research problem is a specific question or issue that requires investigation.

Sources:
- Literature gaps and research papers.
- Real-world and industry problems.
- Emerging technologies.
- Previous projects and academic work.
- Research databases and patents.

Identification Steps: Explore → Narrow → Formulate → Validate.

4.) Good Research Problem:
A good research problem should be:
- Clear and concise.
- Researchable and feasible.
- Relevant and original.
- Suitable for data collection and analysis.

Common Errors:
- Too broad or too narrow
- Unclear problem.
- Lack of resources or time
- Ignoring previous research
- Lack of relevance.

5. Scope and Boundaries

Scope defines the limits of research.

Main Boundaries:

↳ Timeframe
↳ Geographic location
↳ Dataset or target population.
↳ Tools and technologies.
↳ Out-of-scope areas.

Example: Instead of "All in Smart cities", study "Traffic prediction using ML on a selected dataset."

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6.) Research Objectives

Objectives explain ~~the~~ what the research aims to achieve. They provide direction and help define the methodology.

SMART Objectives:

Letter	Meaning
S	Specific — clear
M	Measurable — Can be evaluated
A	Achievable — Realistic
R	Relevant — Related to problem
T	Time-bound — Fixed time

Example: Measure the average latency of an algorithm on a dataset within four weeks.

7.) Data Types

Based on Source:

↳ Primary Data → Collected directly (interviews, sensor readings, experiments).

↳ Secondary Data → Existing information (Kaggle datasets, IEEE articles, government data).

Based on Nature:

↳ Qualitative → Descriptions, opinions, and meanings.

↳ Quantitative → Numerical values and measurements.

8.) Data Collection Tools

Tool	Use
Google Forms	Surveys
Observation Logs	Record behavior
Wireshark	Network packets
Perf Mon	CPU and memory monitoring
Beautiful Soup/Selenium	Web scraping

Choose the tool according to the data required.

9.) Data Analysis and Interpretation

Analysis: Organizing and examining data to find patterns, differences, and relationships.

Interpretation: Explaining what the results mean for the research problem.

Flow: Collect → Analyze → Interpret → Conclude

Remember: A pattern in data does not automatically prove its cause.

10.) Research Instrumentation:
Instrumentation means the hardware and software used for data collection and analysis.

Examples: Sensors; Simulators; Benchmarking tools; Custom AI and network tools.

Calibration: Ensures accurate measurements.

Repeatability: Produces consistent results when an experiment is repeated under the same conditions.

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Research Methodology — Worksheet-2 [Summary]

1.) Dialectic of Research

Dialectic is structured reasoning using different viewpoints to improve a research claim.

Process: Claim → Challenge → Evidence → Refinement.

It helps identify weakness, consider alternatives, and strengthen research arguments.

2.) Models of Arguement

Type	Meaning	Result
Deduction	General → Specific	Certain if premises are true.
Induction	Specific → General	Probable
Abduction	Observation → Explanation	Plausible

→ Deduction: Applies a known rule to a specific case.

→ Induction: Forms a general rule from observations.

→ Abduction: Finds the most likely explanation for an observation.

3.) Proof Methods

Method	Meaning
Demonstration	Shows functionality through examples / prototypes.
Empirical	Uses experiments and real data.
Mathematical	Uses logic and formal proofs

Empirical process : question → Experiment → Measurement → Analysis → Conclusion

4.) Mathematical Proofs :

↳ Direct Proof : Derive the result from unknown assumptions.

↳ Contradiction : Assume the opposite and derive an impossible result.

↳ Induction : Base case → Hypothesis → Inductive step → conclusion.

↳ Contrapositive : Prove $\neg Q \rightarrow \neg P$ to establish $P \rightarrow Q$.

Induction formula :

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Contrapositive :
Original : $P \rightarrow Q$
Contrapositive : $\neg Q \rightarrow \neg P$

Both are logically equivalent.

5.) Deduction vs Induction vs Abduction

↳ Deduction applies known knowledge
↳ Induction generalizes from observations.
↳ Abduction searches for the best explanation.

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6.) Theoretical Models

→ Theoretical Models are abstract representations of computational systems.
→ Examples : Finite Automata, Pushdown Automata, Turing Machines, Graphs, Boolean Algebra, and Formal Grammars.

Turing Machine : Used to study computability and decidability.

Advantages : Clarity, formal analysis, and hardware-independent reasoning.

Limitation : May ignore real-world constraints.

7.) Algorithmic Approaches

Process:

Problem → Algorithm → Complexity → Correctness → Testing.

Includes Time Complexity, space complexity, correctness, and test cases.

Remember: Successful implementation does not guarantee correctness for every input.

8.) Software Engineering Approaches

Common models: Waterfall, Agile, Spiral.

Cycle: Requirements → Design → Implementation → Testing → Maintenance.

Version control, testing, documentation, and reproductibility improve research software.

9.) Mathematical Modelling

Converts a real-world system into a mathematical representation.

Process:

System → Assumptions → Variables → Model → Analysis → Validation.

Queueing: λ = Arrival rate

μ = Service rate

If $\lambda > \mu$, the queue may grow

10.) Performance Evaluation

Measures system effectiveness

Parameter	Meaning
Time	Execution / response time
Space	Memory usage.
Scalability	#workloads Performance with increased #
Reliability	Consistency and failure behaviour

Evaluation cycle : Metric → Experiment → Measurement → comparison → Interpretation.

ML Metrics : Accuracy, Precision, Recall, F1-score, Runtime :

11.) Performance Tool

↳ Profilers : gprof, Valgrind
↳ Simulation : NS3, CloudSim
↳ Benchmarking : SPEC CPU, ML Perf
↳ Monitoring : Prometheus, Grafana.

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Research Methodology — Worksheet-3 [Summary]

1.) Literature Search:

→ Systematic process of finding existing research on a topic.
→ **Purpose**: Understand previous work, find research gaps, avoid duplication.
→ **Sources**: Research papers, books, conference papers, IEEE, ACM, Scopus, Google Scholar.

2.) Literature Review:

→ Study and evaluation of existing research.
→ **Process**: Question → Keywords → Search → Screen → Analyse → Synthesize.

3.) Keywords and Boolean Operators:

Use relevant and specific keys.

↳ AND → narrows search
↳ OR → broadens search
↳ NOT → excludes terms
↳ " " → exact phrase

4.) Citation Chaining:

→ Backward: Check references of a paper.
→ Forward: Find newer papers citing it.

5.) Critical Analysis:

→ Evaluate a paper based on:
Problem → Method → Dataset → results → limitations

Summary tells what the paper says; analysis evaluates it.

6.) Literature Synthesis & Mapping:

Synthesis : Combine findings from multiple papers.

Research Gap : Unexplored or insufficiently studied area.

Mapping: Organizes research by themes, methods, timelines or citations.

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7.) Reference & Citations

→ Give credit to resources and prevent plagiarism.

→ Common styles :

↳ IEE : [1], [2]

↳ APA : (Author, Year)

Tools : Zotero, Mendeley, EndNote.

8.) Plagiarism :

→ Using someone else's words, ideas, code or work without proper credit.

Types: Direct, Mosaic, Self-plagiarism, para-phrasing, Accidental.

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Paraphrasing: Rewrite in your own words and structure + cite the sour

9.) Research Ethics:

Principles for responsible research:

- Honesty
- Objectivity
- Integrity
- Respect
- Accountability

Fabrication: Making up data
Falsification: Manipulating data/ results.

10.) Data Privacy:

- Obtain consent when required.
- Collect necessary data only.
- Protect personal information
- Anonymize data where appropriate.

11.) Authorship:

→ Authorship should be based on meaningful contribution.

- Gift authorship: Credit without meaningful contribution.
- Ghost authorship: Significant contributors not credited.

12.) IRB:

IRB = Institutional Review Board
Reviews research invoving humans to ensure:

Consent + Privacy + Safety + Minimal Risk
+ Risk to Withdraw.

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Research Methodology — [Mid Sem Paper Summary]

Section - A

(Q1.) Steps: Identify problem → literature review → hypothesis → design methodology → collect data → analyze → interpret → report.

(Q2.) Necessity: gives direction / scope, avoids wasted effort, guides methodology, makes objectives measurable.

(Q3.) Plagiarism = using other's work without credit. Avoid via: citation, para-phrasing, quotation marks, plagiarism checkers, original analysis.

(Q4.) Partly agree — research needs fact-finding / evaluation, but also hypothesis formulation and systematic methodology to generate new knowledge.

(Q5.) Literature review = survey of existing research to find gaps.
Effective approach: define scope → search databases → screen quality → synthesize → identify gaps.

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Section – B

Ans 6.) Designs: Randomized Control, Randomized Block, Latin Square, Factorial — each isolates treatment effects from confounds.

Ans 7.) a.) Hypothesis = testable prediction of relationship between variables; purpose is to guide data collection / analysis.

b.) Deductive: theory → specific test.

Inductive → observations → general theory.

Ans 8.) → Fundamental: Knowledge for its own sake, no application.
→ Basic: builds theory, foundation for applied work.
→ Applied: solves specific practical problems.
→ Industrial: applied research for industry product / process improvement.

Ans 9.) Inoculated survival: 15/17 (≈ 88%).
Non-inoculated: 4/14 (≈ 29%).
Inference: vaccine significantly improves survival — effective.

OR: (a) same as Q2.
(b) Modern surveys: online databases, citation tracking, systematic review tools, AI-assisted search.

Section – C

Ans 10.) → Exeperience survey : consult esperienced people to refine problem.

→ Pilot survey : small trial run to test feasibility.

→ Components of research problem : unit/population, variables, context, objectives.

→ Rephrasing : turning vague problems into specific, researchable questions.

~~Ans 11.) a.) Is copying 3 sentences + changing some words + adding citation plagiarism~~

Ans 11.) a.) Yes. Changing a few words but keeping the same sentence structure and ideas is called "patchwriting" — a form of plagiarism. A citation only credits the source ; it doesn't excuse copying its wording. To avoid it, you must fully rewrite in your own words AND cite.

b.) Detection tools scan for matching text strings. Paraphrasing changes the wording / sentence structure, so it won't match word-for-word — lowering the similarity score. But it always stays honest only if you still cite the original source, since the idea still belongs to someone else.

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c.) → May still copy the original's structure/flow too closely (still plagiarism, just harder to detect).

→ Can introduce factual errors or change meaning.

→ Author may not fully understand what they're submitting.

→ Raises the question: whose work is it — the AI's, the original author's, or the student's?

d.) Citation credits who the idea/information originally came from. Paraphrasing only changes the words, not the ownership of the idea. So no — paraphrasing alone never removes the need to cite; you always need both.

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